Simulation for Sustainable Systems

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1 Mock Exam

1.1 Part A

This Petri Net has 4 places and 4 transitions, with only one token at the beginning.
Post Matrix:

$$I^+ = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{1}$$

Pre Matrix:

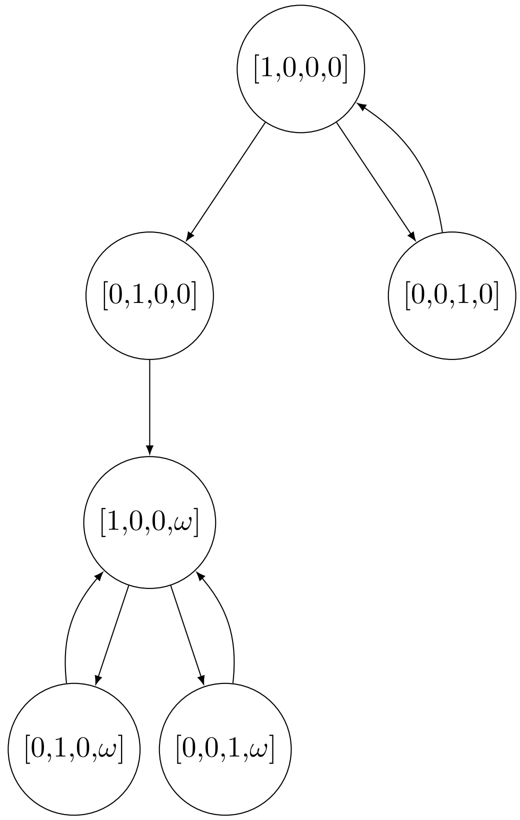
$$I^- \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{2}$$

Incidence Matrix:

$$C = \begin{bmatrix} -1 & -1 & 1 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{3}$$

1.2 Part B

Here I will define $M_0 = (1,0,0,0), M_1 = (0,1,0,0), M_2 = (0,0,1,0), M_3 = (0,0,0,1)$ and use them as shorthand for returns



1.3 Part C

N is not bounded, as a result of p_4, which can consume tokens indefinitely.

1.4 Part D

Completed in Python and verified in Python, the resulting P and T invariant vectors are:

$$P = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad T = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \tag{4}$$

1.5 Part E

$$\mathbf{p}^\top \mathbf{m} = m_1 + m_2 + m_3 = 1 \tag{5}$$

It is a necessary and sufficient condition, as the possible markings reachable from M_0 is (0,1,0,0) and (0,0,1,0).

1.6 Part F

The net does not terminate since there can be infinite firing into p_4.

1.7 Part G

The net is indeed live. From the coverability graph we can see that from any node there is a path to the original position.